Sshubhan Kammari

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Education

University of Manchester — Manchester, UK *Sep 2024 – Jun 2027*
BSc (Hons) Computer Science — Year 2 average: **69% (2:1)**
Relevant modules: Programming Languages & Paradigms, Machine Learning, Software Engineering

Colchester Royal Grammar School — Colchester, UK *Sep 2022 – Jun 2024*
A-levels: Mathematics (A*), Further Mathematics (A), Computer Science (A), Physics (A)

Projects

Low-Latency Market Data Pipeline (C++20, ARM64) | GitHub *Aug 2026 – Oct 2026*

- Built a C++20 pipeline replaying captured Binance futures top-of-book data through a **wait-free SPSC ring buffer**, benchmarked against a tuned mutex/condition-variable queue behind the same interface
- Measured **p99 handoff latency of 125 ns vs 375 ns** for the mutex baseline at 500k msg/s on Apple M2, and showed the ∼12 µs p99.9 floor is time-driven OS scheduling, not per-message cost
- Verified with a sequence-number oracle over **2 billion transfers (none lost, reordered or torn)**, ThreadSanitizer (LLVM, GCC 15) with a negative control that fires, and fault injection on the baseline's tests
- Showed false sharing on M2 disappears at **64 bytes**, below `hw.cachelinesize` (128 B) and libc++ (256 B), and kept 128-byte index alignment as a safety margin

ExamPaper — Practice Paper SaaS (Python, TypeScript) | Live | GitHub *Aug 2026 – Oct 2026*

- Built and deployed a full-stack practice paper app for GCSE and A-level, extensible across subjects and boards and launched with Edexcel GCSE maths (React/TypeScript, **FastAPI**, Postgres, **Playwright**, **Stripe**)
- Designed a **generate-verify pipeline**: seeded question builders derive every answer with **SymPy** and redraw any parameters that fail verification; **250+ tests** cover answers, layout, limits and billing
- Extracted a paper blueprint (topic weightings, mark distributions, question types) from 2022/2023 Edexcel papers, correcting a flat mark distribution to match real papers (**∼21 questions, ∼3.8 marks each**)

PRISM Trading Hackathon — 2nd Place Overall (Python) | GitHub *Apr 2025*

- Placed **2nd overall out of 12+ teams** as a pair; won round 1 (**8,257 pts, +$17,966**), 2nd in round 2
- Built Python bots that polled a REST API for client profiles (budget, age, horizon, disliked sectors) and submitted risk-matched share quantities within **3 s rate limits and 2 s timeouts**
- Adapted to 5 rule changes, from a static high-throughput portfolio to a volatility-ranked, sector-capped allocator with a **5% budget buffer**; prototyped **mean–variance optimisation** (cvxpy)

Experience

Freelance Tutor — Edumentors *Sep 2025 – Present*
Maths, Physics & Computer Science · Remote

- Delivered **235+ one-to-one sessions** of GCSE and A-level tuition in Maths, Physics and Computer Science
- Helped a student pass GCSE Maths after three failed attempts; another rose from grade 3 to 7 in CS

Cloud Computing Work Experience — Capgemini *Jul 2023 (1 week)*
Python, Flask · Remote

- Front-end developer in a 5-person Scrum team building a facial-recognition web app (Flask, Azure Face API); wrote the HTML/Jinja2 interfaces for uploading images and displaying analysis results
- Set up repositories in Azure DevOps and AWS CodeCommit; presented the final solution to Capgemini staff

Technical Skills

- **Languages:** C++20, Python, TypeScript, SQL, JavaScript, C#, HTML/CSS
- **Systems:** Lock-free concurrency, C++ memory model, cache behaviour, latency benchmarking
- **Quant & Data:** Pandas, NumPy, SymPy, cvxpy, Matplotlib
- **Tools:** Git, CMake, Clang/LLVM, GCC, ThreadSanitizer, Linux, macOS, Flask, FastAPI, React, Playwright